

Schedulability Analysis for the Distributed CR-MPC Warehouse Scheduler

1. Problem Setting

We consider a warehouse network modeled as a directed graph $\mathcal{G} = (\mathcal{I}, \mathcal{R}, \mathcal{P})$ where $\mathcal{I}$ is the set of intersections ($|\mathcal{I}| = n_{\text{int}}$), $\mathcal{R}$ the set of road segments ($|\mathcal{R}| = n_{\text{road}}$), and $\mathcal{P}$ the set of external ports (entrances/exits). A fleet of N vehicles is dispatched, each indexed by $n \in \{1, \dots, N\}$ and characterized by an entry port $p_n^{\text{in}} \in \mathcal{P}$, an exit port $p_n^{\text{out}} \in \mathcal{P}$, and a chain of intersections $\mathcal{C}_n \subseteq \mathcal{I}$ traversed en route. We denote $L_n = |\mathcal{C}_n|$ and $\bar{L} = \frac{1}{N} \sum_n L_n$ the average chain length over the fleet.

Each vehicle n has a release time α_n (first arrival at $\mathcal{C}_n[0]$) and an absolute deadline D_n . The control task is to assign per-intersection passage times that respect (i) per-intersection capacity, (ii) per-intersection service rate, and (iii) per-vehicle deadline. We derive sufficient and necessary conditions for the existence of a feasible schedule and the maximum admissible fleet size N_{max} as a function of the controllable design variables.

2. Four Schedulability Bounds

The maximum admissible fleet size is the minimum of four independent bounds, each addressing a distinct feasibility layer:

$$N_{\text{max}} = \min\{N_{\text{cap}}^{(1)}, N_{\text{rate}}^{(2)}, N_{\text{ddl}}^{(3)}, N_{\text{alg}}^{(4)}\}.$$

2.1 Capacity Bound (Multi-commodity Flow with Node Capacity)

Let c_i be the per-intersection capacity (max number of vehicles routed through intersection i within a planning horizon). Each vehicle contributes one unit of demand to every intersection in its chain. The total intersection-visits $\sum_n L_n$ must not exceed network capacity:

$$\sum_{n=1}^N L_n \leq \sum_{i=1}^{n_{\text{int}}} c_i.$$

Assuming uniform capacity $c_i = c$,

$$N_{\text{cap}}^{(1)} = \frac{n_{\text{int}} c}{\bar{L}}.$$

This is the LP relaxation of the multi-commodity flow problem with node capacity constraints, see L  v  que and Lov  sz [1] and Trick [2]. It is a **necessary condition**: any feasible assignment must satisfy it, otherwise the LP itself is infeasible.

2.2 Throughput Bound (Slot-Based Intersection Service Rate)

Each intersection processes vehicles at a service rate $\mu_{\text{int}} = n_{\text{phases}}/T_h$, where n_{phases} is the number of conflict-free vehicle phases that may be granted simultaneously (for a 4-way junction, $n_{\text{phases}} \approx 4$ corresponding to four mutually compatible right-turn movements) and T_h is the per-vehicle clearance time. Over a planning horizon T_{horizon} the total intersection-visits must fit within network throughput:

$$N \cdot \bar{L} \leq n_{\text{int}} \mu_{\text{int}} T_{\text{horizon}},$$

which yields

$$N_{\text{rate}}^{(2)} = \frac{n_{\text{int}} n_{\text{phases}} T_{\text{horizon}}}{T_h \bar{L}}.$$

This bound matches the slot-based intersection capacity analysis of Dresner and Stone [3] and the analytical throughput model of Tachet et al. [4]. The clearance time T_h is in our setting equal to the inter-arrival headway T_{val} enforced at each entrance.

2.3 Deadline Bound (Earliest-Deadline-First with Shared Resources)

Treating each intersection as a unit-capacity preemptive resource, the schedulability of a vehicle set $\{n\}$ with execution requirement C_n (time on the resource) and relative deadline $D_n - \alpha_n$ is governed by the EDF utilization condition (Liu and Layland [5]):

$$\sum_{n=1}^N \frac{C_n}{D_n - \alpha_n} \leq 1.$$

When sharing is non-preemptive, Baker’s Stack Resource Policy [6] introduces a blocking term B and replaces the bound with $\sum_n C_n / (D_n - \alpha_n) \leq 1 - B / D_{\min}$. In our setting C_n is small relative to $D_n - \alpha_n$ (deadlines are computed generously as $D_n = \alpha_n + c_{\text{total}} + D_t \cdot |\mathcal{C}_n - 1|$), hence this bound is rarely binding and can be used as a sanity check rather than a primary design constraint.

2.4 Algorithmic Bound (Decision-Tree Branching Complexity)

The CR-MPC algorithm enumerates feasible passage orderings at each intersection through a decision tree whose worst-case size scales as $\Theta(\prod_i \kappa_i!)$, where κ_i is the number of vehicles sharing intersection i . Pruning rules reduce this empirically but no closed-form upper bound is available. We therefore treat $N_{\text{alg}}^{(4)}$ as an empirically measured ceiling:

Implementation	$N_{\text{alg}}^{(4)}$ on 2×2 grid
MATLAB (with full pruning)	≈ 20
Python port	≈ 12 (boundary 16)

Strengthening pruning is the only way to lift this bound; tuning ADMM hyperparameters does not help.

3. The Effect of Multi-Variant Path Selection

When more than one shortest path connects an OD pair, the generator may select among d equal-length variants. This corresponds to a multi-choice load balancing problem. The classical “balanced allocations” result of Azar, Broder, Karlin, and Upfal [7] gives, for m balls placed sequentially into n bins by selecting the least-loaded among d randomly chosen bins,

$$\max_i \text{load}_i \leq \frac{m}{n} + \Theta\left(\frac{\log \log n}{\log d}\right) \quad (d \geq 2),$$

versus

$$\max_i \text{load}_i \leq \frac{m}{n} + \Theta\left(\sqrt{\frac{m \log n}{n}}\right) \quad (d = 1).$$

In our context $m = N\bar{L}$ and $n = n_{\text{int}}$. With $d = 2$ choices the worst-loaded intersection is much closer to the average load, so the practical capacity utilization $\rho = N\bar{L} / (n_{\text{int}} c)$ at which the generator can still find a feasible assignment rises from $\rho \approx 0.80$ – 0.85 (single variant) to $\rho \approx 0.92$ – 0.95 (two variants). The capacity bound (2.1) thus becomes $N_{\text{cap}}^{(1)} \cdot \rho^*$ where ρ^* depends on d .

For our 2×2 warehouse, of 56 OD pairs 16 (29%) have $d = 2$ equal-length variants, the remaining 40 are single-path. The effective average d is therefore $\bar{d} \approx 1.29$, sufficient to push the achievable fleet from $N \approx 10$ to $N \approx 12$ –13 in our experiments.

4. Application to the 2×2 Warehouse Demo

The reference 2×2 warehouse has $n_{\text{int}} = 4$, $n_{\text{port}} = 8$, intersection layout $\{P_1, P_2, P_3, P_4\}$ at grid corners, and a fully-connected ring of road segments. From OD-pair sampling we measure $\bar{L} = 2.5$. With per-intersection cap $c = 8$, the four bounds evaluate to

$$\begin{aligned} N_{\text{cap}}^{(1)} &= 4 \cdot 8 / 2.5 = \mathbf{12.8}, \\ N_{\text{rate}}^{(2)} &= 4 \cdot 4 \cdot 20 / (2 \cdot 2.5) = 64, \\ N_{\text{ddl}}^{(3)} &\rightarrow \infty \text{ (not binding),} \\ N_{\text{alg}}^{(4)} &= 12 \text{ (Python)} \mid 20 \text{ (MATLAB)}. \end{aligned}$$

Hence

$$N_{\text{max}}^{\text{Python}} = \min(12.8, 64, \infty, 12) = \mathbf{12},$$

$$N_{\text{max}}^{\text{MATLAB}} = \min(12.8, 64, \infty, 20) = \mathbf{12\text{--}13}.$$

The capacity bound (2.1) is the binding constraint for both implementations in the 2×2 setting. Empirical sweep over 17 random seeds with $(N, c) = (12, 8)$ converges 17/17 within a 30 s budget; with $(16, 8)$ the yield drops to 11/17 (65%), consistent with operating at $\rho \approx 1.25$ above the multi-variant safe utilization. With $(20, 8)$ the LP itself is infeasible regardless of seed.

5. Design Levers

The schedulability framework identifies which design variable to manipulate to lift each bound:

Goal	Lever	Bound Affected
Add intersections	$n_{\text{int}} \uparrow$	$N^{(1)}, N^{(2)}$
Loosen density	$c \uparrow$	$N^{(1)}$ (but pressures $N^{(4)}$)
Reduce detours	shorten avg chain via topology	$N^{(1)}$
Add path variants	$d \uparrow$ via topology	utilization ceiling ρ^*
Lengthen horizon	$D_t, T_{\text{ent}} \uparrow$	$N^{(2)}, N^{(3)}$
Faster intersection	$T_h \downarrow$	$N^{(2)}$
Stronger pruning	algorithm	$N^{(4)}$

For the 2×2 warehouse the binding bound is capacity; lifting it requires either reducing $\bar{L}$ (different topology) or accepting higher per-intersection density (which in turn pressures the algorithmic bound). For larger networks ($n_{\text{int}} \geq 6$) the capacity bound recedes and $N_{\text{alg}}^{(4)}$ becomes binding for the Python implementation.

References

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